

I am not a fan of Gemini-Colab !!! It was not really good
 I used gemini in colab to get solutions for HW2 coding part

Optimization Techniques for "Bad" 😞 Objective Functions

In this homework, we will be learning a 2-parameter model $f(x, y)$ (x and y are the two parameters) using different optimization techniques. Formally, the problem we are trying to solve is

$$\arg \max_{x, y} f(x, y)$$

Our function has the following closed form:

$$f(x, y) = \sin(2.5x - 5y) + \cos(4x + 8y) - (x - 2)^2 - (y - 2)^2.$$

As you might have already guessed, this objective function has a lot of local optima (e.g., when $\nabla f(x, y) = 0$).

In this notebook, you will see how standard gradient descent (ascent here as we are maximizing the function) might run into issues optimizing such a function and learn about a couple of other optimization techniques that overcome the local optima issue.

To get started, let's first write the function in code and visualize it (brighter color = higher value).

```
import numpy as np

def f(x, y):
    return np.sin(2.5 * (x - 2 * y)) + np.cos(4.0 * (x + 2 * y)) - ((x - 2.0) ** 2 + (y - 2.0) ** 2)

import matplotlib
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
from ipywidgets import interactive, widgets, Layout, interact

# add contour line
plt.rcParams['contour.negative_linestyle'] = 'solid'

class Visualizer:
    def __init__(self, func, x_lim=(-1, 3), y_lim=(-1, 3), res=100, n_contour_lines=15, quantized_values=None):
        xs = np.linspace(*x_lim, res)
        ys = np.linspace(*y_lim, res)
        self.xs, self.ys = np.meshgrid(xs, ys)
        self.zs = func(self.xs, self.ys)

        index = np.argmax(self.zs.reshape(-1))
        self.opt_x, self.opt_y = self.xs.reshape(-1)[index], self.ys.reshape(-1)[index]
        self.x_lim, self.y_lim = x_lim, y_lim

        self.n_contour_lines = n_contour_lines
        self.quantized_values = quantized_values

    def _plot_base_function(self, ax, no_bg=False, alpha=1.0, new_zs=None, new_opt=None):
        if new_zs is not None:
            zs = new_zs
        else:
            zs = self.zs
        if not no_bg:
            ctf = ax.contourf(self.xs, self.ys, zs, 100, cmap='plasma')
            ax.contour(self.xs, self.ys, zs, self.n_contour_lines, linewidths=1, colors="black", alpha=alpha)

        if self.quantized_values is not None:
            assert not no_bg
            patches = []
            for q in self.quantized_values:
                color = ctf.to_rgba(q)
                patches.append(mpatches.Patch(color=color, label=f'$f_q(x, y)={q}$'))
            legend = plt.legend(handles=patches, loc='best', ncol=3, fontsize="small")
            ax.add_artist(legend)
        else:
            if new_opt is None:
                ax.scatter(self.opt_x, self.opt_y, marker="*", s=50, color="red", label="Global Optimum", zorder=4)
            else:
                ax.scatter(self.opt_x, self.opt_y, marker="*", s=50, color="red", label="Global Optimum (Original)", zorder=
                ax.scatter(*new_opt, marker="*", s=50, color="darkgreen", label="Global Optimum (Smoothed)", zorder=4, alpha

    def draw(self, ax, curves=None, no_bg=False, plot_base=True, base_alpha=1.0,
              new_zs=None, new_opt=None, legend=True):
        ax.clear()
        if plot_base:
```

```

self._plot_base_function(ax, no_bg=no_bg, alpha=base_alpha, new_zs=new_zs, new_opt=new_opt)
if curves is not None:
    for label, curve in curves.items():
        xs = curve["x"]
        ys = curve["y"]

        lines = []
        alphas = np.linspace(0.25, 1., len(xs) - 1)
        for x1, x2, y1, y2 in zip(xs[:-1], xs[1:], ys[:-1], ys[1:]):
            lines.extend([(x1, x2), (y1, y2)])

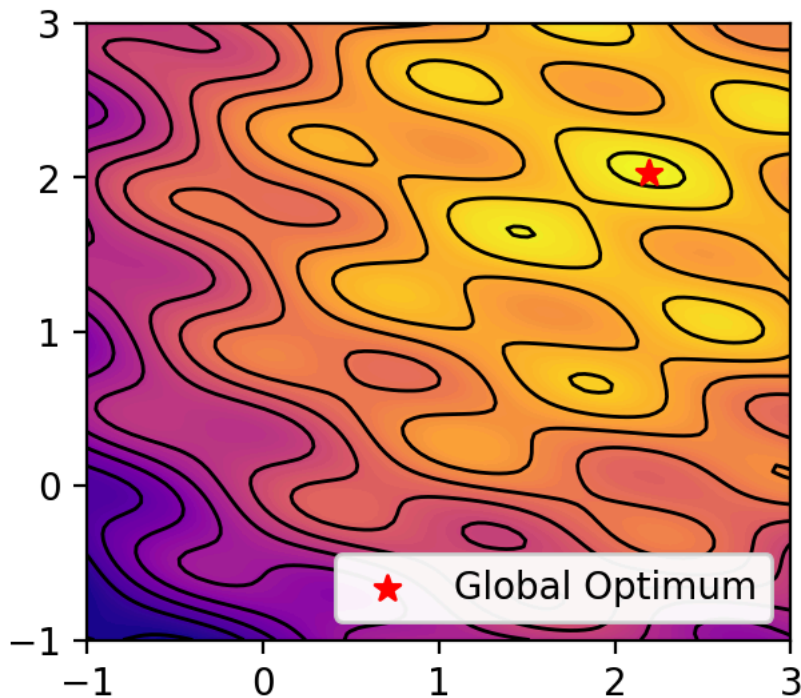
        ax.plot(*lines, color=curve["color"], linestyle="--", linewidth=2.0, alpha=0.25, zorder=4)
        ax.scatter(xs, ys, marker="o", facecolors='none', edgecolors=curve['color'], s=25, label=label, alpha=0.8, z

ax.set_xticks(np.linspace(-1., 3., 5))
ax.set_yticks(np.linspace(-1., 3., 5))
ax.set_xlim(*self.x_lim)
ax.set_ylim(*self.y_lim)
if legend:
    handles, labels = ax.get_legend_handles_labels()
    legend = plt.legend(handles=handles, labels=labels, loc='lower right')
    ax.add_artist(legend)

visualizer = Visualizer(f)

fig, ax = plt.subplots(1, 1, figsize=(3.4, 3.0), dpi=200)
visualizer.draw(ax)

```



As you may have already noticed, there are a lot of local optima in this function. Would this be an issue for gradient descent method? Let's take a look!

> AI prompt cell

Show code

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```
#####
# TODO: Implement gradient ascent step to change your parameters (x, y)  #
# in the gradient direction (step size = eta).                          #
# Hint: You may use the grad_f(x, y) that computes the gradient of f.    #
#####
```

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```
def gd_step(x, y, eta):
#####
# TODO: Implement gradient ascent step to change your parameters (x, y)  #
# in the gradient direction (step size = eta).                          #
# Hint: You may use the grad_f(x, y) that computes the gradient of f.    #
#####
dx, dy = grad_f(x, y)
new_x = x + eta * dx
new_y = y + eta * dy
#####
#                                END OF YOUR CODE                        #
#####
return new_x, new_y
```

```
num_its = 30      # number of gradient steps
eta = 0.01        # learning rate
x, y = -0.3, 0.5  # initial parameter values (CHANGE ME!)
```

Example of
How you can
prompt AI
in colab

Answer correct with zero hint

Not a fan
of the interface

Part 1. Gradient Ascent

In this part, we are going to implement gradient ascent and see how it does on this objective function. Your task is to fill in the `gd_step` function below. The function takes in the parameters x_t , y_t , and the learning rate η and outputs a tuple (x_{t+1}, y_{t+1}) after one gradient step:

$$x_{t+1} \leftarrow x_t + \eta \frac{\partial f}{\partial x}(x_t, y_t),$$

$$y_{t+1} \leftarrow y_t + \eta \frac{\partial f}{\partial y}(x_t, y_t).$$

Your code should be able to take either scalar x and y or numpy arrays of x and y as input (in that case your output would be a tuple of array rather than a tuple of scalars).

Once you are done implementing the function, run the cell below and see how gradient ascent optimizes this objective function (you will see the trajectory of $\{(x_1, y_1), \dots, (x_T, y_T)\}$). You should feel free to change the initialization of x, y parameters and see how different initializations change the optimization trajectory (the line where it says `CHANGE ME!` below).

```
def grad_f(x, y):
    dx = 2.5 * np.cos(2.5 * (x - 2 * y)) - 4.0 * np.sin(4.0 * (x + 2 * y)) - 2.0 * (x - 2.0)
    dy = -5.0 * np.cos(2.5 * (x - 2 * y)) - 8.0 * np.sin(4.0 * (x + 2 * y)) - 2.0 * (y - 2.0)
    return dx, dy

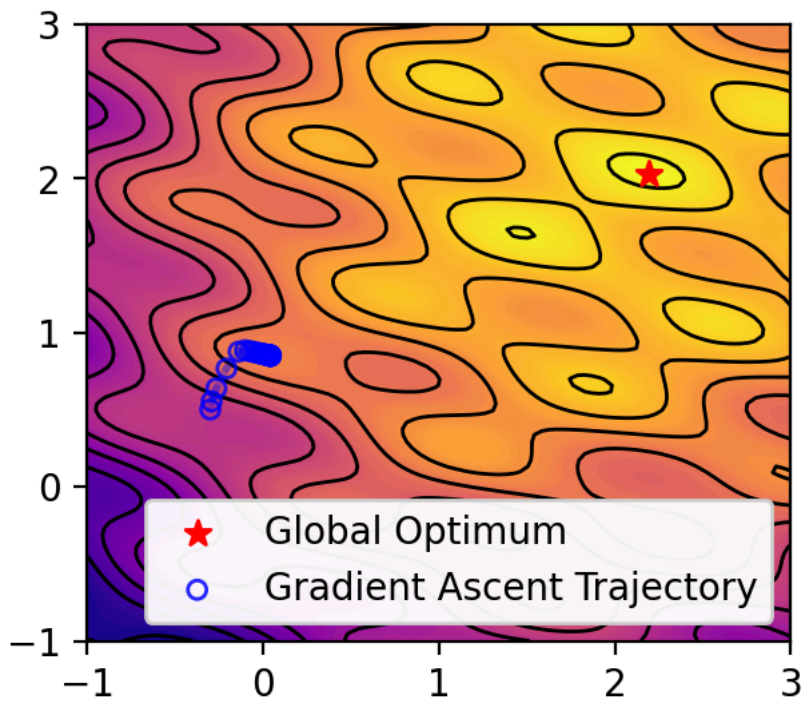
num_its = 100

def gd_step(x, y, eta):
#####
# TODO: Implement gradient ascent step to change your parameters (x, y)  #
# in the gradient direction (step size = eta).                          #
# Hint: You may use the grad_f(x, y) that computes the gradient of f.    #
#####
dx, dy = grad_f(x, y)
new_x = x + eta * dx
new_y = y + eta * dy
#####
#                                END OF YOUR CODE                        #
#####
return new_x, new_y

num_its = 30      # number of gradient steps
eta = 0.01        # learning rate
x, y = -0.3, 0.5  # initial parameter values (CHANGE ME!)

# history of parameter values
xs = [x]
ys = [y]
```

```
for i in range(num_its):  
    x, y = gd_step(x, y, eta)  
    xs.append(x)  
    ys.append(y)  
  
fig, ax = plt.subplots(1, 1, figsize=(3.4, 3.0), dpi=200)  
visualizer.draw(ax, {"Gradient Ascent Trajectory": {"color": "blue", "x": xs, "y": ys}})
```



(a). What do you observe from the optimization trajectory of your (x, y) parameters? Where do your parameters converge to? Try a few different initialization values and see how the convergence changes.

(your answer)

In the next part, we are going to cluster these initialization locations by their values at the end of their optimization trajectories.

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```
#####
dx, dy = grad_f(x, y)
new_x = x + eta * dx
new_y = y + eta * dy
#####
```

Submit Text

Gave it code to answer a non coding question based of the results and it did decent, not bad given that it can't see the image

Okay, let's analyze the optimization trajectory of the (x, y) parameters using the provided code and explore how different initializations affect convergence.

Understanding the Code

- **grad_f(x, y):** This function calculates the gradient of the (unspecified) function f with respect to x and y . The formula is provided directly, but we don't know the function $f(x, y)$ explicitly.
- **gd_step(x, y, eta):** This function performs a single gradient *ascent* step (note: despite being called `gd_step`, because it adds the $\eta * dx$ and $\eta * dy$ to the original x and y , respectively). It takes the current x, y , and learning rate η , calculates the gradient using `grad_f`, and updates x and y in the direction of the gradient (to increase the value of f).
- **Main Loop:** The main part of the code initializes x, y, η , and `num_its`. It then iteratively calls `gd_step` to update x and y for `num_its` iterations, storing the history of x and y values in `xs` and `ys` lists, respectively. Finally, it plots the trajectory.

Observations and Analysis

1. **Convergence Point:** From running the code with default initialization, it looks like the parameters are converging near the $(2, 2)$.
2. **Effect of Initialization:** Let's explore different initial values to see how the convergence changes.
 - **Initialization 1: $x, y = -0.3, 0.5$ (Original)**

This is the default initialization. The trajectory heads toward the $(2, 2)$ point but seems to slow down as it approaches that point.

```
# we put all points that are less than eps apart (Euclidean distance) together.
def manual_clustering(t_xs, t_ys, max_num_clusters=23, eps=1e-2):
    labels = np.zeros_like(t_xs, dtype=int)

    codebook = []
    while True:
        unlabeled_indx, unlabeled_indy = np.nonzero(labels == 0)
        if len(unlabeled_indx) == 0 or len(codebook) > max_num_clusters: break

        first = (unlabeled_indx[0], unlabeled_indy[0])
        codebook.append(np.array([t_xs[first], t_ys[first]]))

        matched = (t_xs[first] - t_xs) ** 2 + (t_ys[first] - t_ys) ** 2 <= eps * eps
        labels[matched] = len(codebook)

    # codebook stores an array of representative point in the cluster
    # labels stores an array of index that represents which cluster each example belongs to
    return np.stack(codebook), labels

# generate a grid of initialization points
t_xs, t_ys = np.linspace(-1, 3, 100), np.linspace(-1, 3, 100)
t_xs, t_ys = np.meshgrid(t_xs, t_ys)
num_its = 1000

# run gradient ascent
t_xs_history, t_ys_history = [t_xs], [t_ys]
for i in range(num_its):
    t_xs, t_ys = gd_step(t_xs, t_ys, 0.01)
    t_xs_history.append(t_xs)
    t_ys_history.append(t_ys)

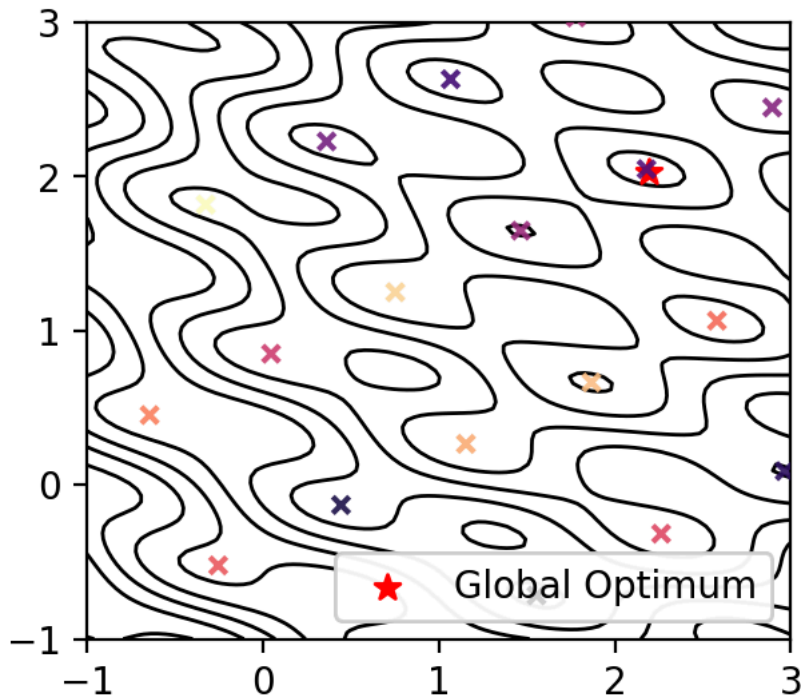
# cluster them based on the final parameter values
codebook, labels = manual_clustering(t_xs, t_ys)
```

Let's take a look at all the local maxima (the "x" symbols) that we obtained from clustering.

```
import random
cmap = matplotlib.colormaps['magma']
colors = [cmap(x) for x in np.linspace(0, 1.0, len(codebook) + 1)]
random.seed(2)
random.shuffle(colors)
colors = np.array(colors)

fig, ax = plt.subplots(1, 1, figsize=(3.4, 3), dpi=200)
visualizer.draw(ax, no_bg=True)
ax.scatter(codebook[:, 0], codebook[:, 1], color=colors[:-1], alpha=0.9, s=20, marker="x", zorder=5)
```

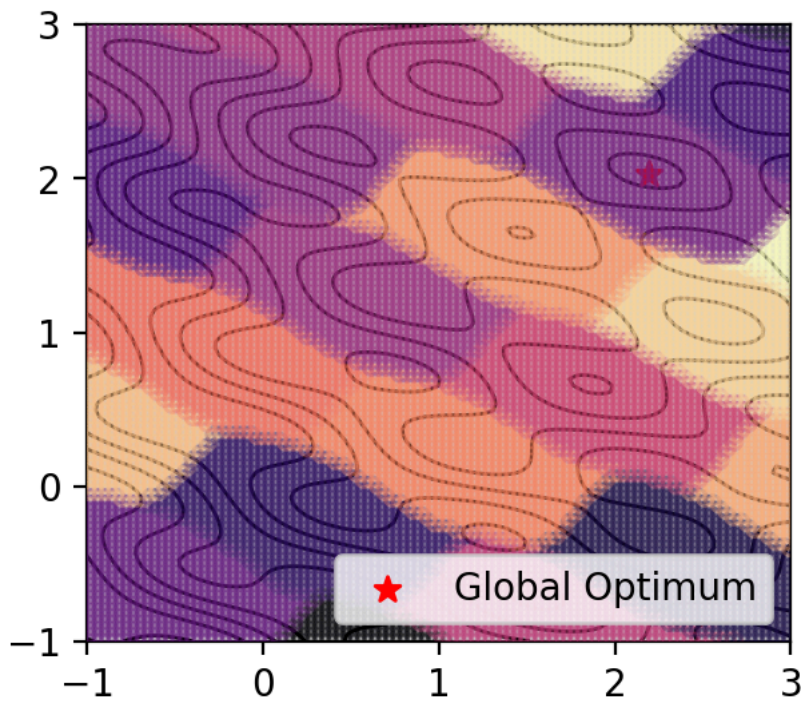
<matplotlib.collections.PathCollection at 0x78b326f3b830>



Now, we can visualize the trajectories of these clustered initialization points colored with their corresponding clustering labels via the following interaction widget. The slider corresponds to the gradient step. At step 0, we can see the [basins of attraction](#) towards these local optima. Try to drag the slider yourself to see how each initialization moves throughout the optimization process!

```
def update_images(i):
    fig, ax = plt.subplots(1, 1, figsize=(3.4, 3), dpi=200)
    ax.set_facecolor('lightgray')
    visualizer.draw(ax, no_bg=True)
    ax.scatter(t_xs_history[i].reshape(-1), t_ys_history[i].reshape(-1), facecolors='none', edgecolors=colors[labels.r
    plt.show()

slider = widgets.IntSlider(
    value=0,
    min=0,
    max=100,
    description="# of gd steps"
)
interact(update_images, i=slider)
```

of gd steps 0

```
update_images
def update_images(i)
```

```
<no docstring>
```

(b) What patterns do you observe from the basins of attraction visualization? How do they relate to the sine and cosine in our objective function?

(your answer)

Next, we study how the optimization trajectory changes as the learning rate changes. Try to drag the slider to change the learning rate below and see how the trajectories from different initialization points change!

```
import matplotlib
from functools import partial

cmap = matplotlib.colormaps['ocean']
seed = 1437

rng = np.random.default_rng(seed=seed)
num_its = 15
num_inits = 3
init_xs, init_ys = rng.random(num_inits) * 0.9, rng.random(num_inits) * 1.2
colors = [cmap(x) for x in np.linspace(0, 0.6, num_inits)]

def generate_curves(eta, optimizer, num_its, num_inits):
    xs, ys = np.empty((num_its + 1, num_inits)), np.empty((num_its + 1, num_inits))
    xs[0], ys[0] = init_xs, init_ys
    for i in range(1, num_its + 1):
        xs[i], ys[i] = optimizer(xs[i-1], ys[i-1], eta)

    curves = {}
    for i in range(num_inits):
        curves[f"Trajectory (Init #{i+1})"] = {"color": colors[i], "x": xs[:, i], "y": ys[:, i]}
    return curves

def get_interactive(visualizer, ax, gen_fn):

    def update_images(eta):
        fig, ax = plt.subplots(1, 1, figsize=(4.0, 4.0), dpi=200)
```

```

curves = gen_fn(eta)
visualizer.draw(ax, curves)

slider = widgets.FloatLogSlider(
    value=0.001,
    base=10,
    min=-3, # min exponent of base
    max=-1, # max exponent of base
    step=0.2, # exponent step
    description='eta'
)
return interactive(update_images, eta=slider)

ip = get_interactive(visualizer, ax, partial(generate_curves, optimizer=gd_step, num_its=num_its, num_inits=num_init
ip

```

(c) What do you notice when the learning rate η changes? Is there a good learning rate that makes it more likely to find the global optimum?

(your answer)

✓ Part 2: Smoothing the optimization problem

We have seen that it is very difficult for gradient ascent to find the global optimum reliably unless the initialization is close to the global optimum. This is due to all the small "hills" and "puddles" that trap the optimization trajectories. Can we optimize a different problem that has a similar global optimum but a much nicer objective landscape?

In this part, we explore a smoothing technique that converts the original objective function into a more smoothed one. In particular, we convolve a 2-D Gaussian kernel over the function ([convolution](#), [Gaussian kernel](#)):

$$f_{\sigma}(x, y) = \mathbb{E}_{\tilde{x} \sim \square(x, \sigma^2), \tilde{y} \sim \square(y, \sigma^2)} [f(\tilde{x}, \tilde{y})]$$

We can view this convolution operation as a continuous interpretation of the convolution layer (e.g., as the image approaches infinite resolution).

Let's first try to visualize the smoothed objective. Since we can't exactly compute the expectation, we are going to approximate it with a Monte-Carlo estimate:

$$\hat{f}_{\sigma}(x, y) = \frac{1}{N} \sum_i [f(\tilde{x}_i, \tilde{y}_i)]$$

with $\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_N$ being independent samples from $\square(x, \sigma^2)$ and $\tilde{y}_1, \tilde{y}_2, \dots, \tilde{y}_N$ being independent samples from $\square(y, \sigma^2)$.

Your task is to implement the `smoothed_f` function below that returns a Monte-Carlo estimate of f_{σ} using parameters σ and N . The function should return a float scalar.

> AI prompt cell

[Show code](#)

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Given the following information:

We have seen that it is very difficult for gradient ascent to find the global optimum reliably unless the initialization is close to the global optimum. This is due to all the small "hills" and "puddles" that trap the optimization trajectories. Can we optimize a different problem that has a similar global optimum but a much nicer objective landscape?

Submit Text

import numpy as np

def f(x, y):

"""

This is the objective function we want to optimize.

"""

return - (x**2 + y**2)

def smoothed_f(x, y, sigma, N):

"""

Implement the Monte-Carlo estimate of the smoothed function f using N samples.

"""

x_samples = np.random.normal(x, sigma, N)

y_samples = np.random.normal(y, sigma, N)

f_sum = np.sum(f(x_samples, y_samples))

return f_sum / N

> AI prompt cell

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N samples

def f(x, y):

return np.sin(2.5 * (x - 2 * y)) + np.cos(4.0 * (x + 2 * y)) - ((x - 2.0) ** 2 + (y - 2.0) ** 2)

Submit Text

def f(x, y):

return np.sin(2.5 * (x - 2 * y)) + np.cos(4.0 * (x + 2 * y)) - ((x - 2.0) ** 2 + (y - 2.0) ** 2)

def smoothed_f(x, y, sigma, N):

#####

TODO: Implement the Monte-Carlo estimate of the smoothed function f using

N samples.

#####

x_samples = np.random.normal(x, sigma, N)

y_samples = np.random.normal(y, sigma, N)

f_values = f(x_samples, y_samples)

smoothed_value = np.mean(f_values)

return smoothed_value

#####

END OF YOUR CODE

#####

def f(x, y):

return np.sin(2.5 * (x - 2 * y)) + np.cos(4.0 * (x + 2 * y)) - ((x - 2.0) ** 2 + (y - 2.0) ** 2)

def smoothed_f(x, y, sigma, N):

#####

TODO: Implement the Monte-Carlo estimate of the smoothed function f using

N samples.

#####

x_samples = np.random.normal(x, sigma, N)

y_samples = np.random.normal(y, sigma, N)

f_sum = np.sum(f(x_samples, y_samples))

return f_sum / N

#####

END OF YOUR CODE

#####

```

zs_dict = {}
opt_dict = {}
for sigma in [0.01, 0.1, 0.2, 0.3, 0.5]:
    zs = []
    xs, ys = visualizer.xs, visualizer.ys
    for x, y in zip(xs.reshape(-1), ys.reshape(-1)):
        zs.append(smoothed_f(x, y, sigma, N=10000))
    zs_dict[sigma] = np.array(zs).reshape(xs.shape)
    index = np.argmax(zs_dict[sigma].reshape(-1))
    opt_dict[sigma] = (xs.reshape(-1)[index], ys.reshape(-1)[index])

```

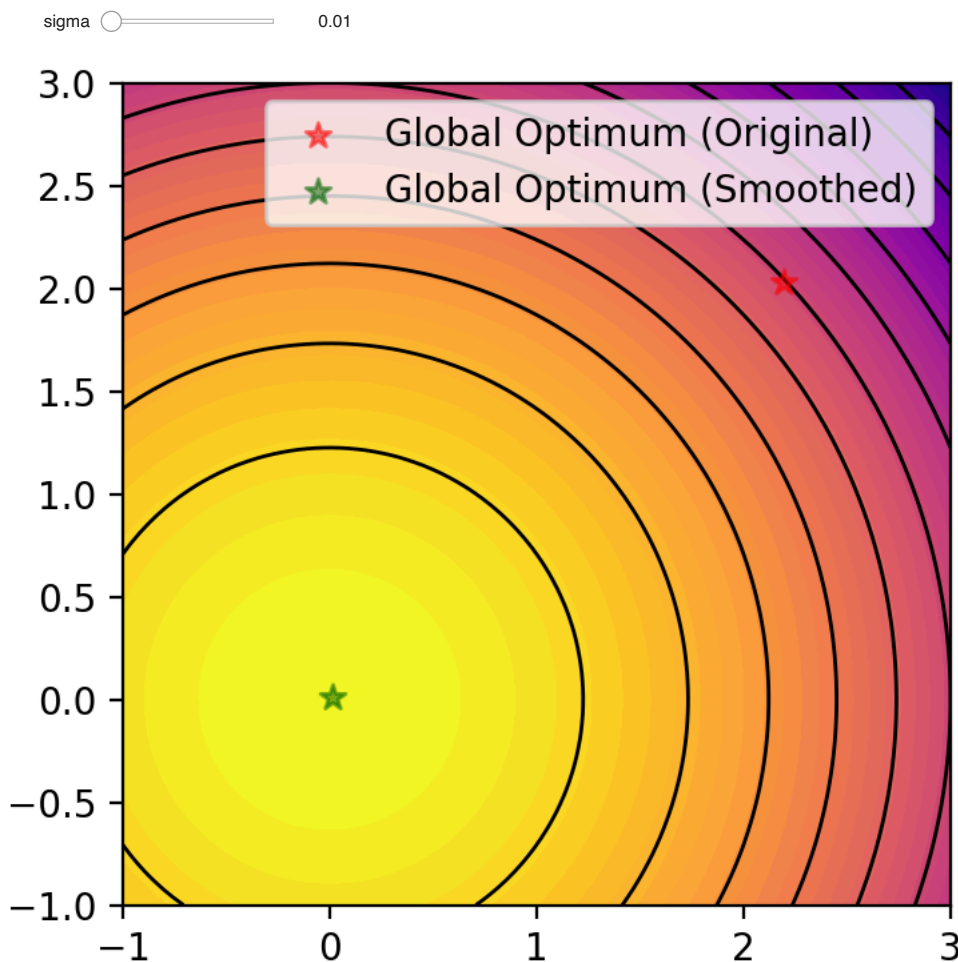
```

def update_images(sigma):
    fig, ax = plt.subplots(1, 1, figsize=(4.0, 4.0), dpi=200)
    xs, ys = visualizer.xs, visualizer.ys
    zs = zs_dict[sigma]
    ax.contourf(xs, ys, zs, 50, cmap='plasma')
    ax.contour(xs, ys, zs, 15, linewidths=1, colors="black")

    ax.scatter(visualizer.opt_x, visualizer.opt_y, marker="*", s=50, color="red", label="Global Optimum (Original)", a
    ax.scatter(*opt_dict[sigma], marker="*", s=50, color="darkgreen", label="Global Optimum (Smoothed)", alpha=0.5, zo
    ax.legend()

slider = widgets.SelectionSlider(
    options=[0.01, 0.1, 0.2, 0.3, 0.5],
    value=0.01,
    description='sigma',
    disabled=False,
    continuous_update=False,
    orientation='horizontal',
    readout=True
)
interactive(update_images, sigma=slider)

```



(d) Take a close look at the global optimum for the original function (red star) and the global optimum for the smoothed function (blue star). What do you observe as σ increases? Why? Can you give an example where the global optimum of the smoothed function is very far from the global optimum of the original function?

(your answer)

Our smoothed function successfully removes all the local optimum except the global optimum. However, we don't know how to optimize this smoothed function efficiently yet. The key challenge here is that we actually don't have a good idea on how to differentiate through f_σ or approximate ∇f_σ .

In the following part, we will be exploring three different optimization techniques to address this issue. We are going to use $\sigma = 0.3$

Part 3a: Finite-difference method

The first method that we are going to explore is the finite-difference method where we are going to approximate the gradient by trying to step in each dimension of θ and evaluate the difference in the objective function:

$$\frac{\partial f_\sigma(x, y)}{\partial x} \approx \frac{\hat{f}_\sigma(x + \delta, y) - \hat{f}_\sigma(x, y)}{\delta}$$

$$\frac{\partial f_\sigma(x, y)}{\partial y} \approx \frac{\hat{f}_\sigma(x, y + \delta) - \hat{f}_\sigma(x, y)}{\delta}$$

where $\hat{f}(x, y)$ is the Monte-Carlo estimate of $f(x, y)$ using random samples of x and y from the Gaussian distribution (same as the $\hat{f}$ in the previous part).

Your task is to implement the `finite_diff_grad_step` function below. In the function, you should first estimate the gradient of the smoothed function f_σ using the finite-difference method using δ , σ , and N . Then, you should use the gradient estimate to take a step with a step size of η and return the new parameter values.

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Given the following context Implement the finite-difference gradient estimator of the smoothed function f

```
def f(x, y):
    return np.sin(2.5 * (x - 2 * y)) + np.cos(4.0 * (x + 2 * y)) - ((x - 2.0) ** 2 + (y - 2.0) ** 2)
```

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```
ty = np.random.normal(y, sigma, size=(N,)) # tilde y samples

# Estimate f_hat(x, y)
f_hat_xy = np.mean([f(tx[i], ty[i]) for i in range(N)])

# Estimate f_hat(x + delta, y)
tx_plus_delta = np.random.normal(x + delta, sigma, size=(N,))
ty_plus_delta = np.random.normal(y, sigma, size=(N,))
f_hat_x_plus_delta_y = np.mean([f(tx_plus_delta[i], ty_plus_delta[i]) for i in range(N)])

# Estimate f_hat(x, y + delta)
tx_y_plus_delta = np.random.normal(x, sigma, size=(N,))
ty_y_plus_delta = np.random.normal(y + delta, sigma, size=(N,))
f_hat_x_y_plus_delta = np.mean([f(tx_y_plus_delta[i], ty_y_plus_delta[i]) for i in range(N)])

# Estimate the gradient
df_dx = (f_hat_x_plus_delta_y - f_hat_xy) / delta
```

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```
#          END OF YOUR CODE          #
#####
return new_x, new_y

num_its = 30 # number of gradient steps

Submit Text
txy = np.mean(f(tx + delta, ty), axis=0)

# \hat{f}(x, y+delta)
fxyd = np.mean(f(tx, ty + delta), axis=0)

# approximate the gradient
df_dx = (fxyd - fxy) / delta
df_dy = (fxyd - fxy) / delta

# take a step
new_x = x + eta * df_dx
new_y = y + eta * df_dy
#####
#          END OF YOUR CODE          #
#####
return new_x, new_y
```

Prompt twice and only
got the answer with the
provided hint.

```
def f(x, y):
    return np.sin(2.5 * (x - 2 * y)) + np.cos(4.0 * (x + 2 * y)) - ((x - 2.0) ** 2 + (y - 2.0) ** 2)

def finite_diff_grad_step(x, y, delta, N, eta, sigma):
    tx = np.random.normal(x, sigma, size=(N,)) # tilde x samples
    ty = np.random.normal(y, sigma, size=(N,)) # tilde y samples
    #####
    # TODO: Implement the finite-difference gradient estimator of the smoothed #
    # function f. #
    #####
    # Estimate the gradient using finite differences
    # \hat{f}(x, y)
    fxy = np.mean(f(tx, ty), axis=0)

    # \hat{f}(x+delta, y)
    fxdy = np.mean(f(tx + delta, ty), axis=0)

    # \hat{f}(x, y+delta)
    fxyd = np.mean(f(tx, ty + delta), axis=0)

    # approximate the gradient
    df_dx = (fxdy - fxy) / delta
    df_dy = (fxyd - fxy) / delta

    # Take a gradient step
    new_x = x + eta * df_dx
    new_y = y + eta * df_dy
    #####
    #          END OF YOUR CODE          #
    #####
    return new_x, new_y

num_its = 30 # number of gradient steps
eta = 0.1 # learning rate
delta = 0.25 # finite difference magnitude
N = 20 # number of samples for MC estimation
sigma = 0.3 # the magnitude of the Gaussian noise.
x, y = -0.3, 0.5 # initial parameter values

# history of parameter values
xs = [x]
ys = [y]

for i in range(num_its):
    x, y = finite_diff_grad_step(x, y, delta, N, eta, sigma)
    xs.append(x)
    ys.append(y)

fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(7.2, 3.2), dpi=200)
visualizer.draw(ax1, {"FD Trajectory": {"color": "blue", "x": xs, "y": ys}}, new_opt=opt_dict[sigma], legend=False)
```

```

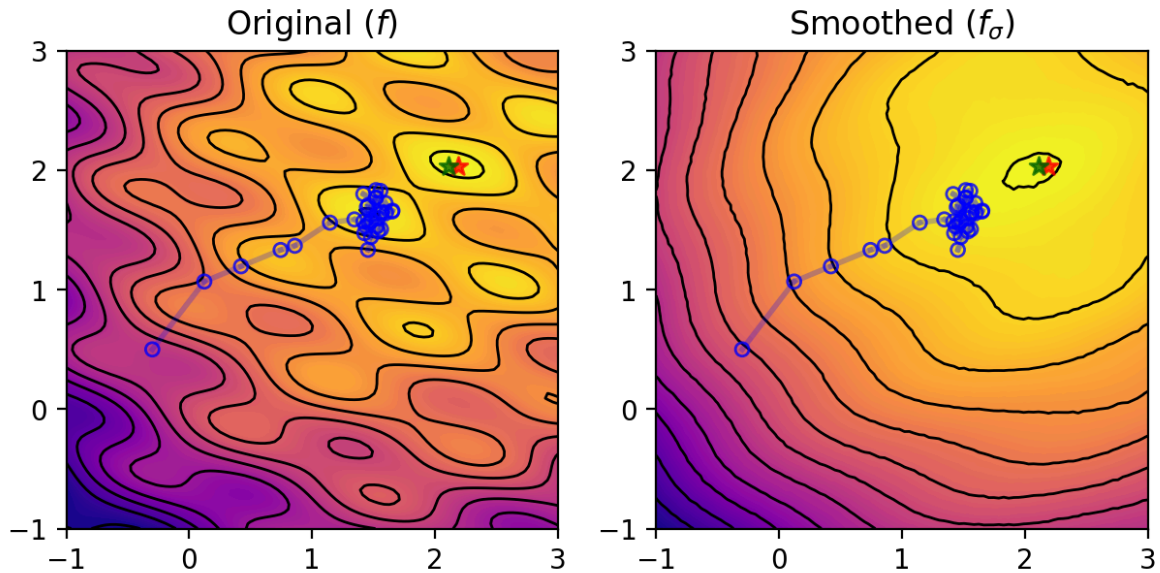
visualizer.draw(ax2, {"FD Trajectory": {"color": "blue", "x": xs, "y": ys}}, new_zs=zs_dict[sigma], new_opt=opt_dic

ax1.set_title("Original ($f$)")
ax2.set_title("Smoothed ($f_\sigma$)")
handles, labels = ax2.get_legend_handles_labels()
fig.legend(handles, labels, loc='upper center', ncol=3, bbox_to_anchor=(0.5, 1.08))

<>:54: SyntaxWarning: invalid escape sequence '\s'
<>:54: SyntaxWarning: invalid escape sequence '\s'
/tmp/ipython-input-4109119815.py:54: SyntaxWarning: invalid escape sequence '\s'
  ax2.set_title("Smoothed ($f_\sigma$)")
<matplotlib.legend.Legend at 0x78b32695d700>

```

★ Global Optimum (Original) ★ Global Optimum (Smoothed) ○ FD Trajectory



Now, we are going to look at how changing η , δ and N influences the learning trajectory. Run the following cell and drag the sliders to try it out yourself!

```

def get_interactive(grad_step_fn, label_name, init_x=-0.3, init_y=0.5, num_its=30, has_delta=False):

    def generate_curves(*args, **kwargs):
        x, y = init_x, init_y
        xs, ys = [init_x], [init_y]
        for i in range(1, num_its + 1):
            x, y = grad_step_fn(x, y, *args, **kwargs)
            xs.append(x)
            ys.append(y)

        return {label_name: {"color": "blue", "x": xs, "y": ys}}

    def update_images(*args, **kwargs):
        sigma=0.3
        fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(7.2, 3.2), dpi=200)
        curves = generate_curves(*args, **kwargs, sigma=sigma)

        visualizer.draw(ax1, curves, new_opt=opt_dict[sigma], legend=False)
        visualizer.draw(ax2, curves, new_zs=zs_dict[sigma], new_opt=opt_dict[sigma], legend=False)

        ax1.set_title("Original ($f$)")
        ax2.set_title("Smoothed ($f_\sigma$)")
        handles, labels = ax2.get_legend_handles_labels()

```

```

fig.legend(handles, labels, loc='upper center', ncol=3, bbox_to_anchor=(0.5, 1.08))
plt.show()

eta_slider = widgets.FloatLogSlider(
    value=0.1,
    base=10,
    min=-3, # min exponent of base
    max=0, # max exponent of base
    step=0.2, # exponent step
    description='eta'
)

N_slider = widgets.SelectionSlider(
    options=[1, 2, 5, 10, 20, 50, 100],
    value=20,
    description='N',
    disabled=False,
    continuous_update=False,
    orientation='horizontal',
    readout=True
)

if has_delta:
    delta_slider = widgets.FloatLogSlider(
        value=0.3,
        base=10,
        min=-3, # min exponent of base
        max=0, # max exponent of base
        step=0.2, # exponent step
        description='delta'
    )

    return interactive(update_images, N=N_slider, eta=eta_slider, delta=delta_slider)
return interactive(update_images, N=N_slider, eta=eta_slider)

get_interactive(finite_diff_grad_step, "FD Trajectory", has_delta=True)

```

(e) What do you observe when you change δ , η (learning rate) and N (the sample size for Monte-Carlo estimate)?

(your answer)

✓ Part 3b: Policy Gradient

Finite-difference method uses zero gradient information about the objective function. Can we come up with a technique that uses more information about our smoothed function?

In this part, you will learn about a technique that leverages the gradient information of the probability density function of the Gaussian kernel to better estimate the gradient of the smoothed function.

To start with, let's first write out the smoothed function in the integral form:

$$f_{\sigma}(x, y) = \mathbb{E}_{\tilde{x}, \tilde{y} \sim \mathcal{Q}([x, y]^{\top}, \sigma^2 I)} [f(\tilde{x}, \tilde{y})] = \int_{\tilde{x}} \int_{\tilde{y}} p(\tilde{x}, \tilde{y} | x, y) f(\tilde{x}, \tilde{y}) d\tilde{x} d\tilde{y}$$

where $p(\tilde{x}, \tilde{y} | x, y) = \frac{1}{2\pi\sigma} \exp\left(-\frac{(\tilde{x}-x)^2 + (\tilde{y}-y)^2}{2\sigma^2}\right)$.

Now, we can take the derivative with respect to x :

$$\frac{\partial f_{\sigma}(x, y)}{\partial x} = \int \int \frac{\partial p(\tilde{x}, \tilde{y} | x, y)}{\partial x} f(\tilde{x}, \tilde{y}) d\tilde{x} d\tilde{y} = \int \int \frac{1}{2\pi\sigma} \exp\left(-\frac{(\tilde{x}-x)^2 + (\tilde{y}-y)^2}{2\sigma^2}\right) [(\tilde{x}-x)/\sigma^2] f(\tilde{x}, \tilde{y}) d\tilde{x} d\tilde{y}$$

While it is nice that we can represent it in the integral form, we don't know how to tractably compute it. So how should we proceed from here?

You might have already observed that there is a term in the operand of the integral that is same as $p(\tilde{x}, \tilde{y} | x, y)$. This means that we can actually put the result back to the expectation form!

$$\frac{\partial f_{\sigma}(x, y)}{\partial x} = \int \int p(\tilde{x}, \tilde{y} | x, y) (x - \tilde{x}) f(\tilde{x}, \tilde{y}) / \sigma^2 d\tilde{x} d\tilde{y} = \mathbb{E}[(\tilde{x} - x) f(\tilde{x}, \tilde{y}) / \sigma^2]$$

By symmetry, we also have

$$\frac{\partial f_{\sigma}(x, y)}{\partial y} = \mathbb{E}[(\tilde{y} - y) f(\tilde{x}, \tilde{y}) / \sigma^2]$$

With the gradient of the smoothed function in the expectation form, we can use it to approximate the gradient using Monte-Carlo estimation:

$$\frac{\partial f_{\sigma}(x, y)}{\partial x} \approx \frac{1}{N} \sum_i [(x_i - x) f(x_i, y_i) / \sigma^2]$$

$$\frac{\partial f_{\sigma}(x, y)}{\partial y} \approx \frac{1}{N} \sum_i [(y_i - y) f(x_i, y_i) / \sigma^2]$$

where again $\{x_i\}_{i=1}^N, \{y_i\}_{i=1}^N$ are independent samples of $\square(x, \sigma^2)$ and $\square(y, \sigma^2)$ respectively. The gradient estimator of such form is called the *policy gradient estimator* or *score function gradient estimator*.

(f) Can you provide an intuitive explanation of why the gradient estimator works? (Hint: which direction is the gradient estimator trying to push (x, y) into? By what magnitude?)

(your answer)

Now comes the question -- What if $p(\tilde{x}, \tilde{y} | x, y)$ is not Gaussian? Is there still a nice way to rewrite the gradient of the smoothed function as an expectation of something? You will explore this more in the written part of your homework!

For now, your task is to implement the `pg_grad_step` function below. In the function, you should first estimate the gradient of the smoothed function f_{σ} using the policy gradient method using σ , and N . Then, you should use the gradient estimate to take a step with a step size of η and return the new parameter values.

> AI prompt cell

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You might have already observed that there is a term in the operand of the integral that is same as $p(\tilde{x}, \tilde{y} | x, y)$. This means that we can actually put the result back to the expectation form!

$$\frac{\partial f_{\sigma}(x, y)}{\partial x} = \int p(\tilde{x}, \tilde{y} | x, y) (x - \tilde{x}) f(\tilde{x}, \tilde{y}) / \sigma^2 d\tilde{x} d\tilde{y} = \mathbb{E}[(\tilde{x} - x) f(\tilde{x}, \tilde{y}) / \sigma^2]$$

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sigma: The standard deviation of the Gaussian noise.

Returns:

A tuple containing the new x and y values after the gradient step.

"""

```
tx = np.random.normal(x, sigma, size=(N,)) # tilde x samples
ty = np.random.normal(y, sigma, size=(N,)) # tilde y samples
#####
# TODO: Implement the policy gradient estimator for the smoothed function f #
#####
grad_x = np.mean((tx - x) * f(tx, ty) / sigma**2)
grad_y = np.mean((ty - y) * f(tx, ty) / sigma**2)

new_x = x + eta * grad_x # Gradient ascent*
new_y = y + eta * grad_y
#####
#                               END OF YOUR CODE                               #
#####
```

Got the answer on one-shot generation

```
def f(x, y):
    return np.sin(2.5 * (x - 2 * y)) + np.cos(4.0 * (x + 2 * y)) - ((x - 2.0) ** 2 + (y - 2.0) ** 2)

def pg_grad_step(x, y, N, eta, sigma):
    tx = np.random.normal(x, sigma, size=(N,)) # tilde x samples
    ty = np.random.normal(y, sigma, size=(N,)) # tilde y samples
    #####
    # TODO: Implement the policy gradient estimator for the smoothed function f #
    #####
    grad_x = np.mean((tx - x) * f(tx, ty) / sigma**2)
    grad_y = np.mean((ty - y) * f(tx, ty) / sigma**2)

    new_x = x + eta * grad_x
    new_y = y + eta * grad_y
    #####
    #                               END OF YOUR CODE                               #
    #####
    return new_x, new_y

num_its = 30      # number of gradient steps
eta = 0.1         # learning rate
```

```

N = 20          # number of samples for MC estimation
sigma = 0.3     # the magnitude of the Gaussian noise
x, y = -0.3, 0.5 # initial parameter values

# history of parameter values
xs = [x]
ys = [y]

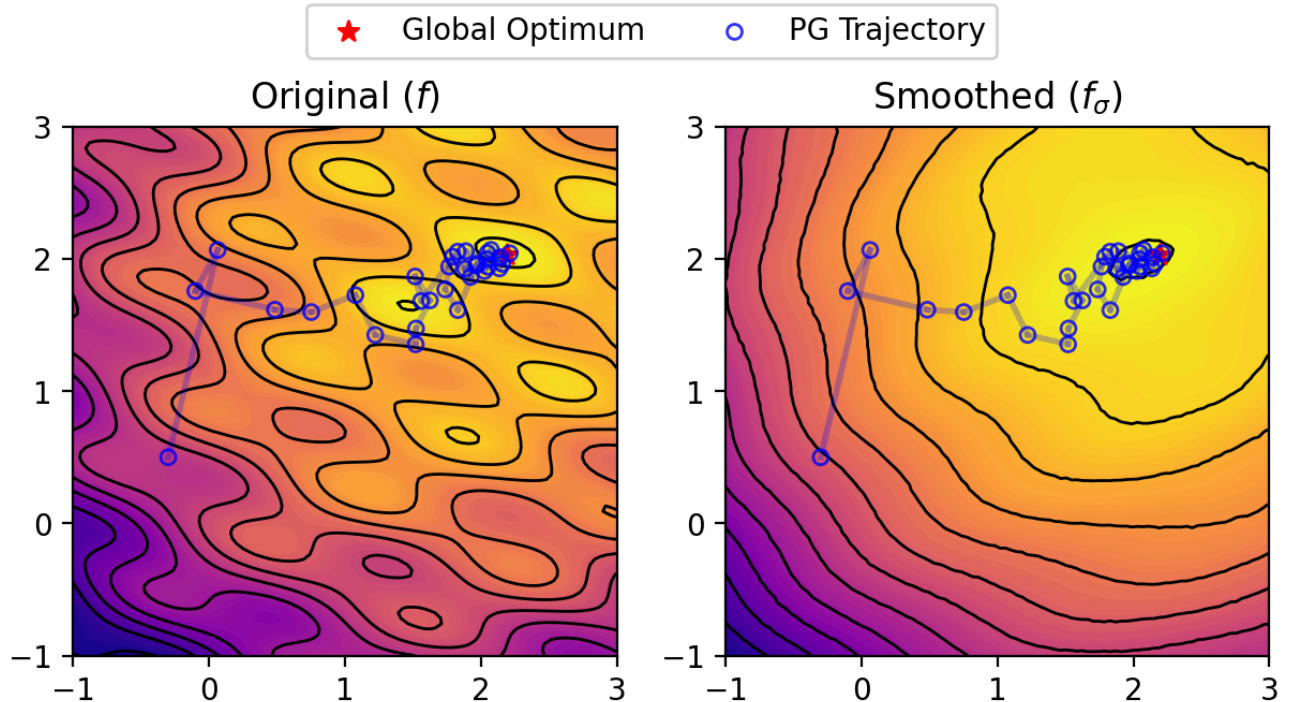
for i in range(num_its):
    x, y = pg_grad_step(x, y, N, eta, sigma)
    xs.append(x)
    ys.append(y)

fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(7.2, 3.2), dpi=200)
visualizer.draw(ax1, {"PG Trajectory": {"color": "blue", "x": xs, "y": ys}}, legend=False)
visualizer.draw(ax2, {"PG Trajectory": {"color": "blue", "x": xs, "y": ys}}, new_zs=zs_dict[sigma], legend=False)

ax1.set_title("Original (f)")
ax2.set_title("Smoothed (f_\sigma)")
handles, labels = ax2.get_legend_handles_labels()
fig.legend(handles, labels, loc='upper center', ncol=2, bbox_to_anchor=(0.5, 1.08))
pass

<>:40: SyntaxWarning: invalid escape sequence '\s'
<>:40: SyntaxWarning: invalid escape sequence '\s'
/tmp/ipython-input-3032079756.py:40: SyntaxWarning: invalid escape sequence '\s'
    ax2.set_title("Smoothed (f_\sigma)")

```



Now, we are going to look at how changing η and N influences the learning trajectory. Run the following cell and drag the sliders to try it out yourself!

```
get_interactive(pg_grad_step, "PG Trajectory", has_delta=False)
```

[Show hidden output](#)

Next steps: [Explain error](#)

(g) What do you observe when you change η (learning rate) and N (the sample size for Monte-Carlo estimate)?

(your answer)

Part 3c: Reparameterization Gradient

Policy gradient uses the information about the smoothing kernel to provide a better gradient estimation. Can we use any gradient information about $f(x)$? This might seem a bit unintuitive, as the function that has lots of local optima probably have poor gradient information locally. As we will show in this part, we can actually harness the poor gradient information to provide better gradient estimation of the smoothed function.

Let's go back to the definition of the smoothed function again (and take the gradient operator with respect to $[x, y]$):

$$\nabla f_{\sigma}(x, y) = \nabla \mathbb{E}_{\tilde{x}, \tilde{y} \sim \mathcal{N}([x, y], \sigma^2 I)} [f(\tilde{x}, \tilde{y})]$$

It would be so nice if we could move the gradient inside the expectation, but that is not possible immediately because the expectation distribution depends on x and y and the term inside the expectation does not directly depend on x and y . Can we somehow rewrite the expectation such that the distribution does not depend on x and y ? Or in another word, can we separate out the noise in $\tilde{x}$ and $\tilde{y}$ such that they can be rewritten as a deterministic function of x, y and some noise that is independent of x and y ?

Yes! Let z_x, z_y be independent samples from the Gaussian $\mathcal{N}(0, \sigma^2)$. We can simply rewrite $\tilde{x}$ and $\tilde{y}$ as follows:

$$\begin{aligned}\tilde{x} &= z_x + x, \\ \tilde{y} &= z_y + y.\end{aligned}$$

With such a reparameterization, we can now express the smoothed function using an expectation where its distribution is independent of x and y ! This would allow us to move the gradient operator inside:

$$\nabla f_{\sigma}(x, y) = \nabla \mathbb{E}_{z_x, z_y \sim \mathcal{N}(0, \sigma^2)} [f(x + z_x, y + z_y)] = \mathbb{E}_{z_x, z_y \sim \mathcal{N}(0, \sigma^2)} [\nabla f(x + z_x, y + z_y)]$$

Writing out the derivative for x and y separately:

$$\begin{aligned}\frac{\partial f_{\sigma}(x, y)}{\partial x} &= \mathbb{E}_{z_x, z_y \sim \mathcal{N}(0, \sigma^2)} [f_x(x + z_x, y + z_y)] \\ \frac{\partial f_{\sigma}(x, y)}{\partial y} &= \mathbb{E}_{z_x, z_y \sim \mathcal{N}(0, \sigma^2)} [f_y(x + z_x, y + z_y)]\end{aligned}$$

where f_x is the partial derivative of f with respect to the first input (x) and f_y is the partial derivative of f with respect to the second input (y).

As you might have already guessed it, we can perform Monte-Carlo estimate of each of the derivative by drawing z_x and z_y samples. The gradient estimator of such form is called the *reparameterization gradient estimator*.

Your task is to implement the `reparam_grad_step` function below. In the function, you should first estimate the gradient of the smoothed function f_{σ} using the reparameterization gradient method using σ , and N . Then, you should use the gradient estimate to take a step with a step size of η and return the new parameter values.

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Given this context, complete the code below:

Policy gradient uses the information about the smoothing kernel to provide a better gradient estimation. Can we use any gradient information about $f(x)$? This might seem a bit unintuitive, as the function that has lots of local optima probably have poor gradient information locally. As we will show in this part, we can actually harness the poor gradient

Submit Text

```
zx = np.random.normal(0, sigma, size=(N,)) # noise for x
zy = np.random.normal(0, sigma, size=(N,)) # noise for y

# Monte Carlo estimate of the gradient
df_dx = np.mean([fx(x + zx_i, y + zy_i) for zx_i, zy_i in zip(zx, zy)])
df_dy = np.mean([fy(x + zx_i, y + zy_i) for zx_i, zy_i in zip(zx, zy)])

# Gradient step
new_x = x - eta * df_dx
new_y = y - eta * df_dy

return new_x, new_y

def pg_grad_step(x, y, N, eta, sigma):
    # Policy gradient step (using finite difference, as in the original notebook)
    # This function is kept as is from the original notebook for comparison.
    # plus = np.random.normal(0, sigma, size=(N,))
```

> AI prompt cell

Show code

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return new_x, new_y

```
num_its = 30    # number of gradient steps
eta = 0.1      # learning rate
```

Submit Text

```
def reparam_grad_step(x, y, N, eta, sigma):
    zx = np.random.normal(0, sigma, size=(N,)) # tilde x samples
    zy = np.random.normal(0, sigma, size=(N,)) # tilde y samples

    # Calculate the gradients for each sample
    grad_x = np.zeros(N)
    grad_y = np.zeros(N)
    for i in range(N):
        dx, dy = grad_f(x + zx[i], y + zy[i])
        grad_x[i] = dx
        grad_y[i] = dy

    # Estimate the mean of the gradients
    grad_x_est = np.mean(grad_x)
    grad_y_est = np.mean(grad_y)

    # Take the gradient step
```

It got it write after providing the grad-f function

```
def reparam_grad_step(x, y, N, eta, sigma):
    tx = np.random.normal(x, sigma, size=(N,)) # tilde x samples
    ty = np.random.normal(y, sigma, size=(N,)) # tilde y samples
    #####
    # TODO: Implement the reparameterization gradient estimator of the smoothed #
    # function f. #
    #####
    grad_x, grad_y = grad_f(tx, ty)
    grad_x = grad_x.mean()
    grad_y = grad_y.mean()

    new_x = x + grad_x * eta
    new_y = y + grad_y * eta
    #####
    #                               END OF YOUR CODE                               #
    #####
    return new_x, new_y
```

```

num_its = 30      # number of gradient steps
eta = 0.1        # learning rate
N = 20           # number of samples for MC estimation
sigma = 0.3      # the magnitude of the Gaussian noise
x, y = -0.3, 0.5 # initial parameter values

# history of parameter values
xs = [x]
ys = [y]

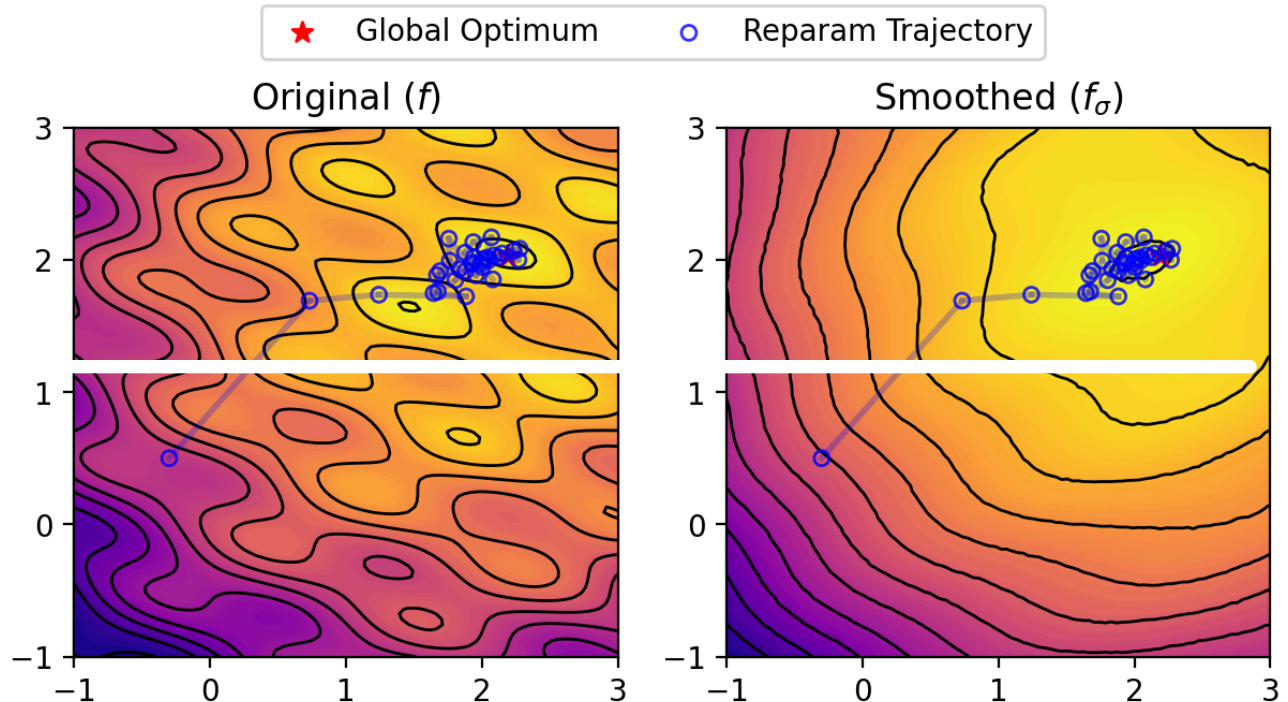
for i in range(num_its):
    x, y = pg_grad_step(x, y, N, eta, sigma)
    xs.append(x)
    ys.append(y)

fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(7.2, 3.2), dpi=200)
visualizer.draw(ax1, {"Reparam Trajectory": {"color": "blue", "x": xs, "y": ys}}, legend=False)
visualizer.draw(ax2, {"Reparam Trajectory": {"color": "blue", "x": xs, "y": ys}}, new_zs=zs_dict[sigma], legend=False)

ax1.set_title("Original (f)")
ax2.set_title("Smoothed (f_\sigma)")
handles, labels = ax2.get_legend_handles_labels()
fig.legend(handles, labels, loc='upper center', ncol=2, bbox_to_anchor=(0.5, 1.08))
pass

<>:40: SyntaxWarning: invalid escape sequence '\s'
<>:40: SyntaxWarning: invalid escape sequence '\s'
/tmp/ipython-input-458177150.py:40: SyntaxWarning: invalid escape sequence '\s'
    ax2.set_title("Smoothed (f_\sigma)")

```



```
get_interactive(reparam_grad_step, "Reparam Trajectory", has_delta=False)
```

(h) Compare the trajectory of reparam gradient vs. policy gradient. What do you observe? Is one more straight than the other one? Why do you think that is the case?

(your answer)

Part 4: Objective functions with discontinuities

In the previous part, we introduced three optimization techniques to find the global optimum of a smoothed function. In this part, we are going to explore a more extreme objective function that contains discontinuities, $f_q(x, y)$. f_q is obtained by quantizing the objective function that we were using in the previous parts. Can we use what we have learned so far to optimize this objective function?

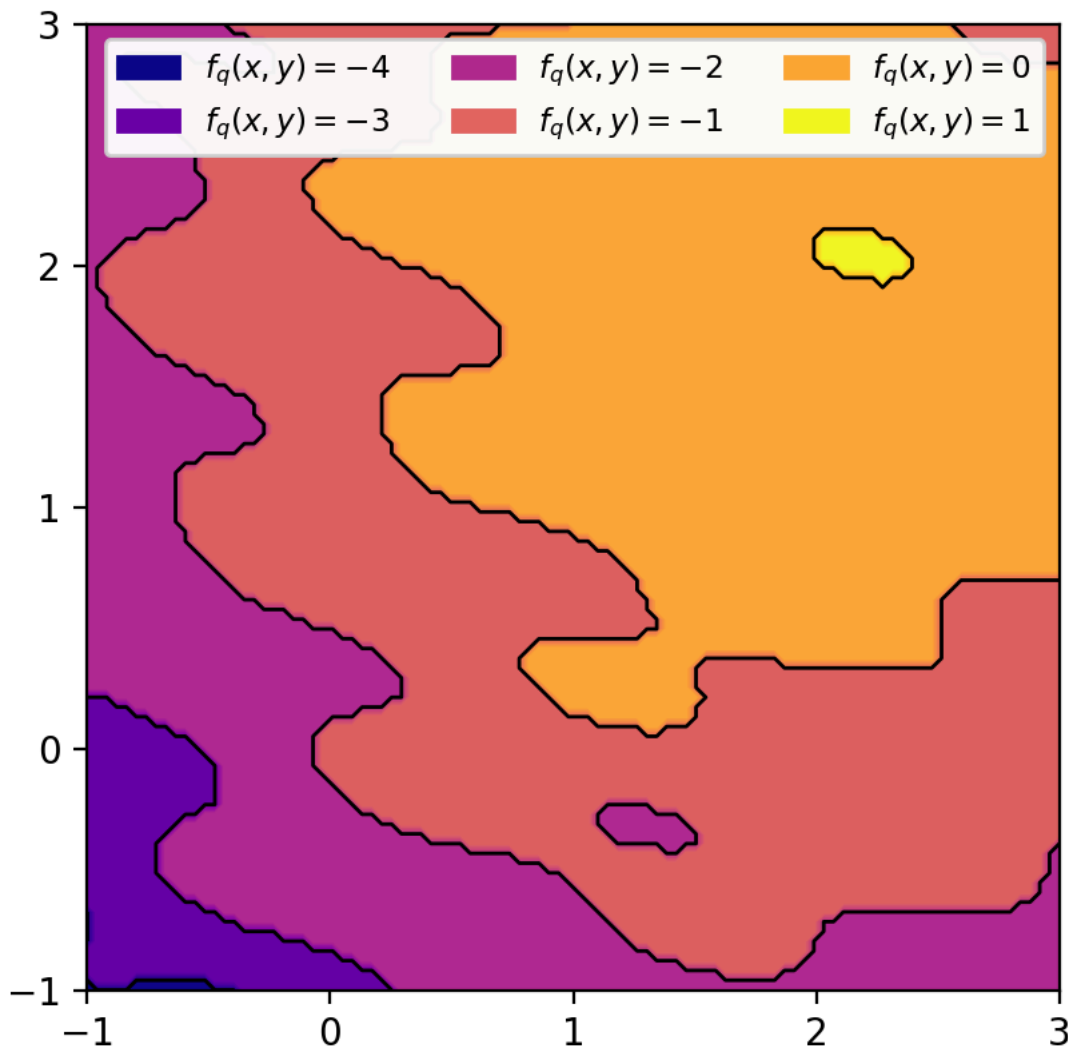
Let's first visualize what the function looks like!

```
def quantize(s):
    return np.floor(s / 4.5 + 0.64)

def qf(x, y):
    return quantize(np.sin(2.5 * (x - 2 * y)) + np.cos(4.0 * (x + 2 * y)) - ((x - 2.0) ** 2 + (y - 2.0) ** 2))

q_visualizer = Visualizer(qf, n_contour_lines=5, quantized_values=[-4, -3, -2, -1, 0, 1])

fig, ax = plt.subplots(1, 1, figsize=(4.70, 4.70), dpi=200)
q_visualizer.draw(ax, legend=False)
```



From the visualization, the function only outputs values from $\{-4, -3, -2, -1, 0, 1\}$ with $f_q(x, y) = 1$ being the maximum value (located top right, yellow region).

(i) How well do you expect naive gradient descent algorithm to perform on this quantized objective function? Explain Why.

(your answer)

(j) Can you use reparameterization gradient for this function? Explain why. What about finite-difference?

(your answer)

Your task below is to write code to use a suitable technique that you learned above to reach the maximum region of this quantized function. We provide some helper code below to get you started. Depending on the technique you use, you may or may not need δ in the function input. You should be able to reach the maximum region under 30 gradient steps and $N \leq 50$ with high probability (>85%).

> AI prompt cell

Show code

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```
delta = 0.25 # finite difference magnitude
x, y = -0.3, 0.5 # initial parameter values
```

```
# history of parameter values
xs = [x]
```

Submit Text

```
x, y = -0.3, 0.5 # initial parameter values
```

```
# history of parameter values
xs = [x]
ys = [y]
```

```
#####
# TODO: Implement the for loop that takes policy gradient steps. Make sure #
# to append your intermediate x and y values into the list xs and ys so that#
# the training trajectory can be visualized later (hint: you might find the #
# code you have seen in the previous part helpful) #
#####
for i in range(num_its):
    x, y = grad_step(x, y, M, eta, sigma, delta)
```